



**Question Bank**  
**Subject: Engineering Physics**  
**Course: B.tech**  
**Semester: 1st**

**PART A – Short question**

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| 1   | Compute Significance of Plank's Constant?                                              |
| 2   | State the Heisenberg uncertainty principle?                                            |
| 3   | Define de-Broglie hypothesis?                                                          |
| 4   | What is Ultraviolet catastrophe? Explain?                                              |
| 5   | What do you mean by Compton effect?                                                    |
| 6   | Discuss the salient features of free electron theory.                                  |
| 7   | Illustrate the V-I characteristics of a p-n Junction diode?                            |
| 8   | Explain the terms amplitude, frequency and phase of a wave.                            |
| 9   | Compare Simple harmonic motion with oscillatory motion?                                |
| 10  | Discuss Main Components of a Laser                                                     |
| 11  | What are the three distinct processes by which a transition can take place?            |
| 12  | Write Maxwell equations in Differential form.                                          |
| 13  | Give the physical interpretation of Differential form of Gauss's law in magnetostatics |
| 14. | Explain de-Broglie hypothesis.                                                         |
| 15. | Write the physical significance of a wavefunction?                                     |
| 16  | Interpret direct and indirect bandgaps in semiconductors?                              |
| 17  | Discuss types of semiconductors?                                                       |
| 18  | What do you mean by SHM?                                                               |
| 19  | Differentiate between the damped and undamped vibrations                               |
| 20  | Write Difference between Interference and Diffraction                                  |
| 21  | What do you mean by population inversion?                                              |
| 22  | Give the importance of displacement current.                                           |

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| 23 | What do you understand by electric field and electric potential |
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## PARTB- Long question

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| 1  | Explain the effective mass of electron and concept of hole.                                                                                                                                  |
| 2  | Explain the origin of energy bands in solids. On the basis of band theory how the crystalline solids are classified into metals, semiconductors and Insulators?                              |
| 3  | Explain Brillouin zones with the help of E-K diagram                                                                                                                                         |
| 4  | Explain the significance of Fermi energy level. Mention its position in intrinsic and extrinsic semiconductors at 0 K?                                                                       |
| 5  | Calculate the de Broglie wavelength for (a) a proton of kinetic energy 70 MeV kinetic energy and (b) a 100 g bullet moving at $900 \text{ m s}^{-1}$ ?                                       |
| 6  | Discuss Wave–particle Duality? The de-Broglie wavelength of the electron is $30\text{\AA}$ . Calculate the phase velocity and group velocity of the matter wave associated with the electron |
| 7  | Classify Conductors, Semi conductors and Insulators on basis of Band theory of solid?                                                                                                        |
| 8  | How does the Fermi level play a significant role in semiconductors?                                                                                                                          |
| 9  | Define simple harmonic motion? Derive the differential equation for SHM from Displacement                                                                                                    |
| 10 | Write the characteristics of a laser beam? Explain Spontaneous Emission and Stimulated Emission?                                                                                             |
| 11 | Discuss the necessary conditions for obtaining sustained interference                                                                                                                        |
| 12 | Distinguish between Fresnel's and Fraunhofer's diffraction                                                                                                                                   |
| 13 | Discuss Equation of Continuity?                                                                                                                                                              |
| 14 | Explain Deferential form of Faraday's law?                                                                                                                                                   |
| 15 | What do you mean by Compton effect? Explain?                                                                                                                                                 |
| 16 | Calculate the smallest possible uncertainty in the position of an electron moving with velocity $3 \times 10^7 \text{ m/s}$                                                                  |
| 17 | Differentiate intrinsic semiconductor and extrinsic semiconductor with example.                                                                                                              |
| 18 | Generate an expression for the concentration of electrons in conduction band in Intrinsic semiconductors.                                                                                    |
| 19 | Why Simple harmonic motion is called sinusoidal Justify it?                                                                                                                                  |
| 20 | A particle executes S.H.M of period 10 sec and amplitude 5.0 cm. Determine the Maximum amplitude of velocity?                                                                                |
| 21 | What is interference of light? Discuss the necessary conditions for obtaining Sustained interference                                                                                         |
| 22 | Discuss Main Components of a Laser?                                                                                                                                                          |
| 23 | Explain the construction, principle and working of a ruby laser?                                                                                                                             |

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| 24 | Explain Biot–Severt law ?Find electric Field Intensity due Long Straight Conductor?                                               |
| 25 | Write Four Maxwell's equation. Explain their Physical Significance?                                                               |
| 26 | Prove that Velocity of electromagnetic Wave in Conducting Medium is less than the velocity of Electromagnetic wave in free Space. |

**PARTC-Very Long question**

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| 1  | Explain quantum mechanical Tunnelling? Derive Expression for the energy in a rectangular potential well.                                                                                                                                 |
| 2. | Calculate the Compton shift if X-rays of wavelength $1.0 \text{ \AA}$ are scattered from a carbon block. The scattered radiation is viewed at $90^\circ$ to the incident beam.                                                           |
| 3  | What are the salient features of quantum free electron theory? What are the merits and demerits?                                                                                                                                         |
| 4  | How does current flows in a PN junction diode? State some application of PN junction diode?                                                                                                                                              |
| 5  | What do you mean by matter wave? If an electron is subjected to a potential difference $V$ volts then prove that the corresponding de Broglie wavelength is $\lambda = (12.26/\sqrt{V}) \text{ \AA}$ .                                   |
| 6  | Solve the problem of a particle free to move within a potential box with rigid walls extending from $x=0$ to $x=a$ . Prove that the energy is given by $E_n = \hbar^2 n^2 \pi^2 / 2ma^2$ . Plot the wave functions for $n=1,2$ and $3$ . |
| 7  | Write the significance of Fermi energy level. Establish its position in intrinsic and extrinsic semiconductors at $0 \text{ K}$ .                                                                                                        |
| 8  | An intrinsic semiconductor has an energy gap of $0.4 \text{ eV}$ . Calculate the probability Of occupation of the lowest level in conduction band at $50^\circ\text{C}$ .                                                                |
| 9  | Explain the terms amplitude, frequency and phase of a wave? Derive the differential equation for SHM from energy principle                                                                                                               |
| 10 | Derive a relation between Einstein's A&B coefficients.                                                                                                                                                                                   |
| 11 | In Newton's ring experiment, the diameter of the $10^{\text{th}}$ dark ring is $0.433\text{cm}$ . Find the wavelength of incident light, if the radius of curvature of the lens is $70 \text{ cm}$ .                                     |
| 12 | Explain Displacement Current?                                                                                                                                                                                                            |
| 13 | Prove that Velocity of Electromagnetic wave in Free Space is equal to velocity of light?                                                                                                                                                 |
| 14 | Drive Expression for the Planck's Formula of Black Body Radiation?                                                                                                                                                                       |
| 15 | Calculate the work function in electron volts of a metal, given that photoelectric Threshold is $6200 \text{ \AA}$                                                                                                                       |
| 16 | State and proved Heisenberg uncertainty principle ?Write some application of Heisenberg uncertainty principle                                                                                                                            |
| 17 | Explain E-K Diagram? Draw E-K Diagram for Ge and Si.                                                                                                                                                                                     |
| 18 | Derive an expression for the concentration of electrons in conduction band and Holes in valence band?                                                                                                                                    |
| 19 | Calculate the number of donor atoms per $\text{m}^3$ of $n$ -type material having resistivity Of $0.25 \text{ }\Omega\text{-m}$ , the mobility of electrons is $0.3 \text{ m}^2/\text{v-s}$ .                                            |
| 20 | Derive the differential equation for the damped natural oscillation of a particle? Give the conditions for over damping, critical damping and under damping action.                                                                      |
| 21 | The amplitude of an oscillator of frequency $200$ per second falls to $1/10$ of its Initial value after $2000$ cycles. Calculate (i) relaxation time, (ii) quality factor                                                                |

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| 22 | Give the description of He-Ne laser? Explain the construction and working of He-Ne Laser? |
| 23 | Derive Modified form of Ampere's Circuital law?                                           |
| 24 | Prove that electromagnetic wave is transverse in nature?                                  |